VERONICA EYO

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Results-driven Data Scientist with 5+ years of experience delivering production AI, machine learning, NLP, predictive modeling, Generative AI, and agentic AI solutions. Proven ability to translate complex business challenges into scalable, production-ready solutions and measurable outcomes across the full ML lifecycle. Combines advanced analytical expertise with strong business acumen, technical leadership, and cross-functional collaboration across insurance, engineering, and technology.

Brings 10+ additional years of engineering and technology experience, including 9+ years leading electrical and instrumentation projects across complex oil and gas EPC, DED, and FEED environments, along with 2 years in systems engineering, network management, IT development, implementation, and support.

TECHNICAL SKILLS, CORE COMPETENCIES & OPERATIONAL STRENGTHS

Programming & Data: Python, SQL, R, C, C++, C#, PySpark, Pandas, NumPy, PostgreSQL, MySQL, SQL Server, Azure SQL, DB2, Snowflake
Machine Learning & AI: Machine Learning, CatBoost, XGBoost, LightGBM, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Generative AI, Computer Vision, Text Mining, Feature Engineering, Model Training & Evaluation, SHAP, Predictive Modeling, Model Monitoring, Data & Model Drift Analysis, Model Versioning
Generative AI & Agentic Systems: OpenAI, Azure OpenAI, MCP (Model Context Protocol), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, AI Agents, Prompt Engineering, Gradio, TruLens, Flowise, n8n, SmartScraperGraph
Document Intelligence & Retrieval: OCR, Document Processing, Information Extraction, Unstructured Data Processing, LLM-Based Document Analysis, Email & Loss-Run Feature Extraction, PDFPlumber, PyMuPDF, Tesseract, Vector Search, Vector Databases, Semantic Retrieval
Cloud, MLOps & Data Engineering: Microsoft Azure, Azure Machine Learning, Azure SQL, Azure Data Factory, Azure DevOps, Azure Container Instances, Docker, AWS SageMaker, Amazon S3, CloudWatch, ETL/ELT Pipelines, Production ML Pipelines, Production Scoring Pipelines, Model Deployment, Model Monitoring, Model Versioning, Data Pipeline Development
Analytics & Business Intelligence: Power BI, Tableau, Looker, Apache Superset, Google Analytics, Data Analytics, Data Visualization, Business Intelligence, Statistical Analysis, Exploratory Data Analysis, Microsoft Excel
Engineering Tools: MATLAB, AutoCAD, SmartPlant Instrumentation (INtools), SolidWorks
Core Strengths: End-to-End ML Solution Development, Production AI/ML Deployment, Agentic AI Solutions, Healthcare Informatics, Insurance & Underwriting Analytics, Model Performance Analysis, Model Governance & Monitoring, Cross-Functional Collaboration, Technical Communication, Stakeholder Engagement, Strategic Problem Solving, Mentorship & Training, Engineering & Project Delivery

EDUCATION

Master of Science in Electrical and Computer Engineering, University of Florida, Gainesville, Florida
Bachelor of Engineering – Electrical and Computer Engineering, Federal University of Technology Minna, Niger State, Nigeria

CERTIFICATIONS

Generative AI with Large Language Models, Google Data Analytics Professional, IBM certification for Python in Data Science.

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PROFESSIONAL EXPERIENCE

Data Scientist- Machine Learning & AI Solutions: Skyward Specialty Insurance, Mar 2024-Present

- Architected, productionized, and monitored machine-learning submission triage solutions across multiple E&S insurance programs including ESPR, ESXS/ESXU, and MEO, integrating structured underwriting data with signals extracted from emails, loss runs, and submission documents to prioritize accounts by estimated bind likelihood. Improved underwriting efficiency by 35%, with dynamic quantile-based prioritization projected to support up to $3.5M in incremental 2026 revenue.
- Led development and production enhancement of the ESPR underwriting prioritization model, building CatBoost-based scoring pipelines that classify submissions into actionable A/B/C/D tiers while incorporating eligibility rules, hard filters, renewal logic, dynamic thresholds, and auditable score reasons.
- Designed an explicit missing-email modeling strategy that expanded scoring coverage to submissions without usable email correspondence, incorporating email-availability signals and regex-derived features and enabling model-variant analysis across more than 54,000 submissions.
- Built model monitoring and drift-analysis frameworks to evaluate training-to-production feature shifts, population stability, tier migration, operational filtering, and model-version behavior, providing actionable insights into production model performance and governance.
- Conducted deep-dive underwriting outcome analysis linking model tiers, filter results, quote activity, and bind outcomes; identified 190 filtered submissions that ultimately bound, including 147 ETF-only accounts and demonstrated that nearly 80% were renewals, directly informing changes to production scoring logic.
- Engineered and migrated production scoring infrastructure to Azure SQL/MDSM, designing intake, extraction, feature, and model data layers and implementing reproducible Python/SQL workflows for scoring, model versioning, threshold management, validation, and auditability.

- Developed scalable NLP and document-intelligence pipelines using OCR, regex, RAG, and LLM-based extraction to transform unstructured broker correspondence, loss runs, claims notes, and insurance documents into structured features and analytical datasets for downstream ML applications.
- Delivered an agentic MCP server-client analytics assistant called the EDAbot that translates natural-language business questions into governed database queries and generates downloadable Excel and PDF outputs, enabling self-service analytics for business users.
- Automated complex insurance document workflows, including reinsurance treaty analysis, claims extraction, competitor intelligence, retention analysis, and underwriting correspondence analysis using RAG, LangChain, OpenAI/Azure OpenAI, PyMuPDF, PDFPlumber, Tesseract, and related document-intelligence technologies.
- Deployed and operationalized ML and AI workflows using Azure Machine Learning, Azure Data Factory, Azure DevOps, Azure SQL, and Docker, supporting end-to-end model deployment, data integration, production scoring, monitoring, and lifecycle management.
- Partner with underwriters, actuarial, data engineering, business leaders, and technology teams to translate underwriting requirements into production AI/ML solutions, communicate complex model findings to nontechnical stakeholders, and drive data-informed business decisions; mentor analysts in RAG, AI/ML, and analytical methodologies.
- Developed and deployed a DistilBERT model with TensorFlow and PyTorch to identify key date references and terms in insurance PDFs, enhancing our underlying data for RAG systems.
- Automated claims data extraction and high-value claims flagging from loss run PDFs using a RAG-based approach with PDFPlumber, PyMuPDF, Tesseract OCR, and GPT-3.5; visualized results in Power BI for actionable insurance insights.
- Designed and executed hypothesis-driven underwriting and pricing analyses using statistical testing and predictive modeling, generating insights tied to revenue, profitability, submission conversion, claims behavior, and underwriting strategy.
- Created and executed comprehensive **test plans** to validate data integrity and completeness from SQL-based data pipelines, ensuring reliability of inputs used across modeling workflows.
- Designed and executed hypothesis-driven R&D initiatives that optimized underwriting strategy, leading to 3 months projected $344K revenue gain through improved submission status conversions.
- Applied statistical methods including t-tests and chi-square tests to validate hypotheses and assess significance of observed trends in claims, pricing, and underwriting data.
- Leveraged predictive modeling with CatBoost to simulate the impact of high-performing underwriter pricing strategies across peer groups; analysis revealed that replicating quoting strategy alone in mid-market segments could reduce profitability by $500K, underscoring the importance of deeper strategic drivers beyond surface-level pricing behavior.
- Developed a competitor intelligence RAG system using SmartScraperGraph, GPT-4o, and LangChain to automate email classification and data extraction for more targeted underwriting and marketing campaigns.
- Utilized a RAG framework to analyze competitor data from non-bound broker-underwriter correspondence, developing a loss reason classification system and predictive claims frequency model using LightGBM, presented in Power BI.
- Streamlined wrongful act claims analysis by implementing a RAG solution to extract key acts and dates from unstructured claim notes using LLMs (GPT-4o-mini) and LangChain.
- Created a RAG pipeline using LangChain, GPT-4o-mini, and PyMuPDF to analyze incumbent carrier behavior, resulting in a retention reason classification system and predictive price elasticity model.
- Supervised and mentored Data Analysts on implementing RAG methodologies, , contributing to skill development and team growth.

Senior Healthcare Data Scientist (3-Month Contract): Equity Quotient, Jan 2025-Apr 2025

- Completed a short-term contract supporting data-driven strategy for healthcare organizations by analyzing large-scale CMS datasets (Medicare, Medicaid) using advanced statistical modelling and machine learning techniques (e.g., XGBoost, SHAP).
- Delivered actionable insights on cost-effectiveness, quality metrics, and health equity through robust modeling and evaluation, ensuring compliance with CMS, HIPAA, and data governance standards.
- Created impactful visualizations and reports using Python, Tableau, and Power BI to communicate findings across both technical and non-technical stakeholders.

AI Analyst: Nabafat AI, Dec 2022-Jan 2024

- Built and deployed Retrieval-Augmented Generation (RAG) systems for dynamic PDF interaction and real-time customer service automation using LangChain, OpenAI, and Gradio.
- Fine-tuned Large Language Models (LLMs) for domain-specific applications, enhancing accuracy and contextual relevance in both customer-facing and internal use cases.

- Created a suite of Generative AI tools using Hugging Face models for tasks such as text summarization, image captioning, image generation, and LLM chat interfaces—all showcased via Gradio demos.
- Deployed Hugging Face models on AWS SageMaker and containerized RAG evaluation workflows using TruLens with Docker to benchmark and improve LLM performance.
- Led end-to-end development of an automated ordering assistant that operated 24/7, integrating LLMs with user workflows for enhanced responsiveness and reliability.

Data Science Instructor, Bloom Institute of Technology, Sep 2022- Mar 2024

- Taught Data Science concepts to students via live online sessions with code-alongs, real-life examples, and storytelling.
- Provided individualized coaching, reviewed assignments, and delivered constructive feedback to Data Science students.
- Mentored and supported learners by providing encouragement based on their progress.
- Provided curriculum feedback and reported on learner progress to school authorities

Data Scientist: Evolent Health, Feb 2022-Jan 2023

- Mentored and onboarded interns and new team members on problem-solving methodologies, company policies, and efficient tool utilization.
- Conducted Exploratory Data Analysis, Data Wrangling, and Data Transformation using MS Power BI on data from MySQL, PostgreSQL, and SAS databases for a Client Churn Analysis Project.
- Automated the extraction of contract text and signature detection from client PDF files using Tabula, YoloE, and Tesseract.
- Analyzed data from primary and secondary sources like AWS S3 buckets using PySpark, Python, and R to deliver actionable insights and trend analyses to management.
- Developed, validated, and implemented machine learning models on clinical datasets.
- Applied NLP libraries (Spacy, NLTK, BERTopic) and techniques to analyze company member reviews and generate insightful topic models.
- Generated weekly reports and real-time dashboards using Apache Superset, and Tableau.
- Utilized AWS SageMaker and Microsoft Azure for repository management and model deployment.
- Conducted research on emerging technologies and industry trends to upgrade existing solutions and proactively apply new opportunities to business challenges.

Select Highlight:

- Implemented an automated contract text extraction and signature detection system from client PDFs (Tabula, YoloE, Tesseract), resulting in $1.2 million cost saving for the company in 2022 by eliminating the need for additional manual staff.

Career Note: From 2021 to 2022 focused on the successful completion of Data Science program with Innovatics.

Senior Electrical Engineer: DUOS TECHNOLOGIES INC · Apr 2020-Apr 2021

- Designed Electrical, Instrumentation, and Control Schematics, Instrument Location Drawings, and Installation Details.
- Led engineering teams in developing Scopes, Engineering sketches, Drawings, and Specifications into complete packages.
- Developed new Test Procedures and performed electrical testing of systems to ensure proper function and client requirement fulfillment.
- Prepared Cable Schedules according to customer standards and NEC regulations.
- Conducted post-construction electrical field inspections to verify compliance with contract documents.
- Interfaced with vendors to evaluate new products and specifications for integration into new package designs.
- Implemented Ethernet-based Local Area Networks and other communication systems.

Select Highlights:

- Designed and deployed the innovative Oblique VUE system for undercarriage train inspection (a first-of-its-kind solution).
- Successfully developed and designed new inspection perspectives that were retrofitted into existing systems and deployed based on customer specifications.

Senior Instrumentation and Controls Engineer: NIGERIAN NATIONAL PETROLEUM CORPORATION SUBSIDIARY, NETCO (NNPC-NETCO) - Aug 2012- Dec 2019

- Ensured timely delivery of major FEED, DED, and EPC projects for clients including NGPTC, NPDC, and NGC, from conceptual design through to As-Built documentation.
- Oversaw the development of comprehensive Instrumentation and Control deliverables, including Design Philosophies, Datasheets, Loop Diagrams, Hook-up Diagrams, Layout Plans, Instrument Installation Details, Junction box Termination Diagrams, Cable block Diagrams, Control System Block Diagrams, Material Take-Offs, and Requisition packages.

- Directed proficiency in precise Relief Valve, Control Valve, and Restriction Orifice sizing using specialized software (Crosby, Daniel Orifico) and ensured team mastery of Smart Plant Instrumentation (INtools) and AutoCAD.
- Facilitated seamless cross-disciplinary collaboration with other engineering teams to ensure integrated project design and execution.

Select Highlights:

- Successfully completed assigned tasks effectively, surpassing time frame objectives while maintaining superior levels of performance.

Career Note: From 2009 to 2011 focused on the successful completion of Master of Electrical and Computer Engineering Degree.

Systems Engineer: ALLIED COMPUTERS LIMITED, Abuja, Nigeria- Feb 2007- Jul 2009 .

- Key contributor to support team, serving as systems engineer focused on delivering top quality hardware / software support to government organizations.
- Carried out expert evaluation and installation of software, hardware, and equipment into client networks
- Facilitated troubleshooting and maintenance efforts by other team members working on computer and printer systems with faulty components.
- Provided supervision and mentoring of junior engineering staff as well as generated weekly performance evaluation reports focused on engineering team.

Select Highlights:

- Upgraded over 500 systems from Windows XP to Windows Vista in one week using only intern resources, successfully delivering the project through strategic resource management and full lifecycle oversight.
- Spearheaded the deployment of Point of Sale (POS) Terminals at various locations.

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AFFILIATIONS

Affiliations: Institute of Electrical and Electronics Engineers (IEEE), National Society for Black Engineer - UF, Women in Electrical & Computer Engineering - UF, Society of Women Engineers (SWE) - UF